FORTH
INSTITUTE OF COMPUTER SCIENCE

Evaluating HPK for Running Cloud-Native Workloads on Slurm Clusters

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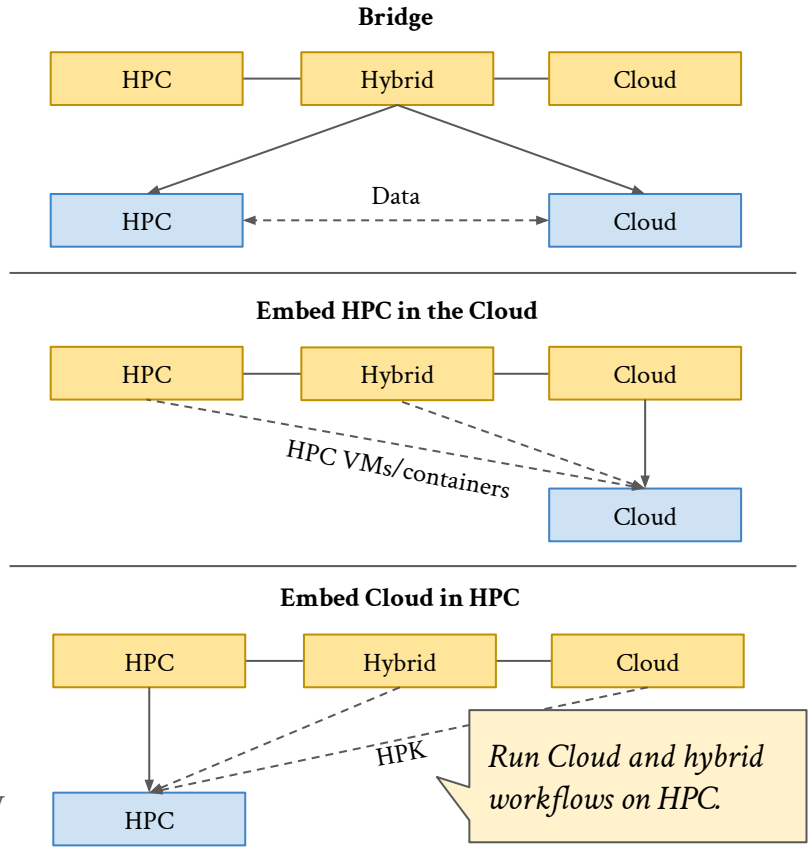
Running hybrid workloads

- Bridge the two separate environments
  - Submit HPC jobs from the Cloud side or vice versa[1]
  - Deal with separate data and network contexts, unless running HPC in the Cloud (VM offerings available)
  - Hardware and maintenance costs
- Run both on the same hardware
  - Synchronize schedulers or use either for scheduling
- Run Slurm in Kubernetes
  - Embed the HPC software stack in containers[2]
- Run Kubernetes in Slurm with HPK!
  - Create ephemeral Kubernetes mini-Clouds in HPC[3]
  - Why? → EuroHPC, stability/consistency, multi-tenancy



[1] KNoC is a Kubernetes node to manage container lifecycle on HPC clusters: https://github.com/CARV-ICS-FORTH/knoc (InteractiveHPC 2022)
[2] Genisys is a Kubernetes scheduler for running HPC jobs inside Virtual Clusters alongside other services: https://github.com/CARV-ICS-FORTH/genisys (VHPC'22)
[3] HPK allows users to run Kubernetes in Slurm: https://github.com/CARV-ICS-FORTH/HPK (WOCC'23)

Design goals

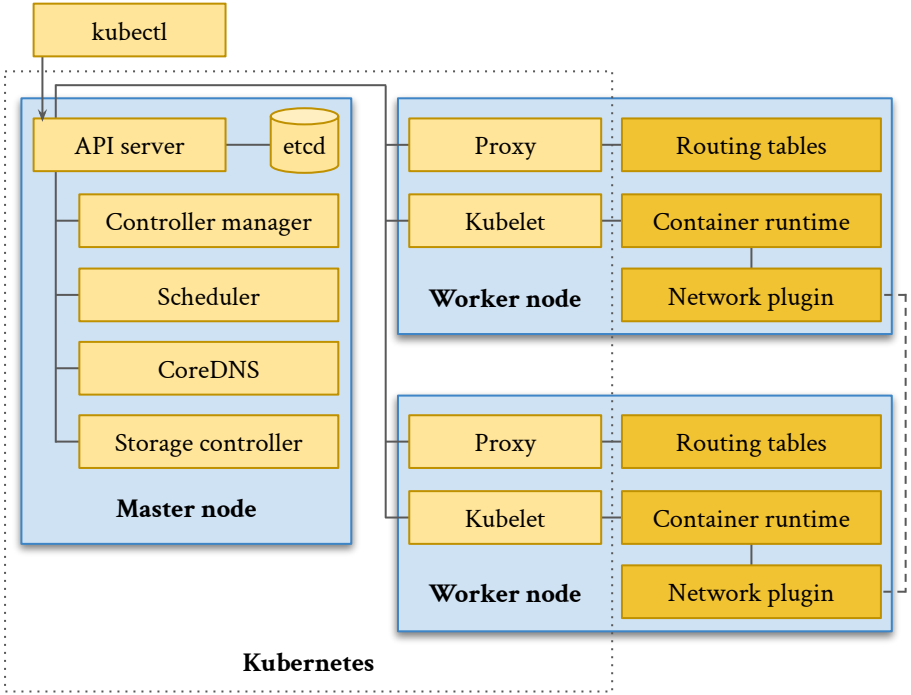
HPK → Run *Kubernetes in an HPC environment* as a user

Requirements:

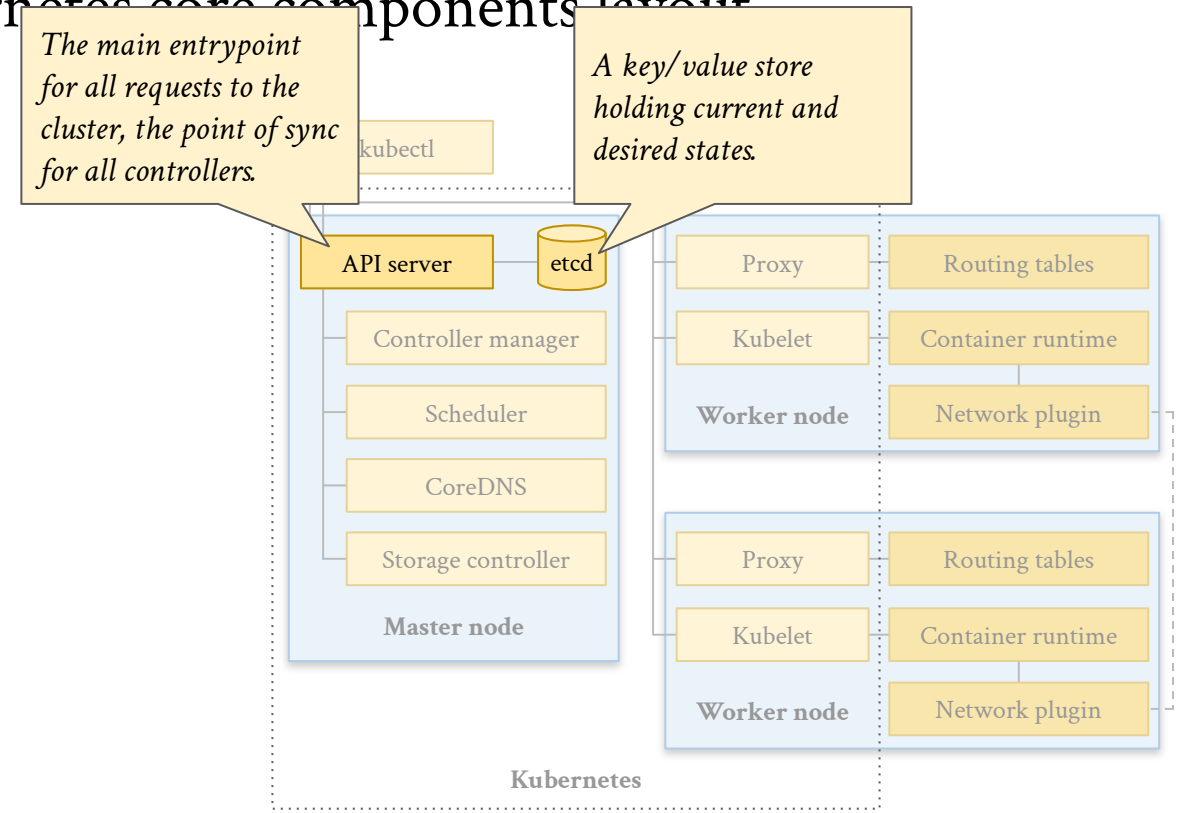
- All* Kubernetes abstractions should be available and fully functional
  - *Except those that relate to physical hardware resources (like NodePort services)
  - Private, inter-container network and internal DNS should work as expected
- Delegate resource management to Slurm
  - Respect organization policies
  - Comply with established resource accounting
  - Scale across all nodes of the cluster
- Use Singularity as the container runtime
  - Preinstalled in most HPC environments
- Make it easy for HPC administrators
  - No (or little) configuration changes should be required at the host level
  - No reliance on special libraries or binaries that execute with "elevated" permissions
- Make it easy for users
  - All neatly packaged up in one container
  - Simple, one-command deployment via Slurm
  - All relevant configuration and files in $HOME

Kubernetes technical concepts

- Single API endpoint
- Multiple controllers/operators
- Objects stored in etcd
  - Pods → Collection of containers
  - Deployments → Replicated pod groups
  - Services → Microservice naming
  - Jobs → Pods that run to completion
  - Volumes → Mountable file collections
- Typical distributed structure
  - "Control plane" → Scheduling and placement
  - Node agents → Handle execution



Kubernetes core components layout



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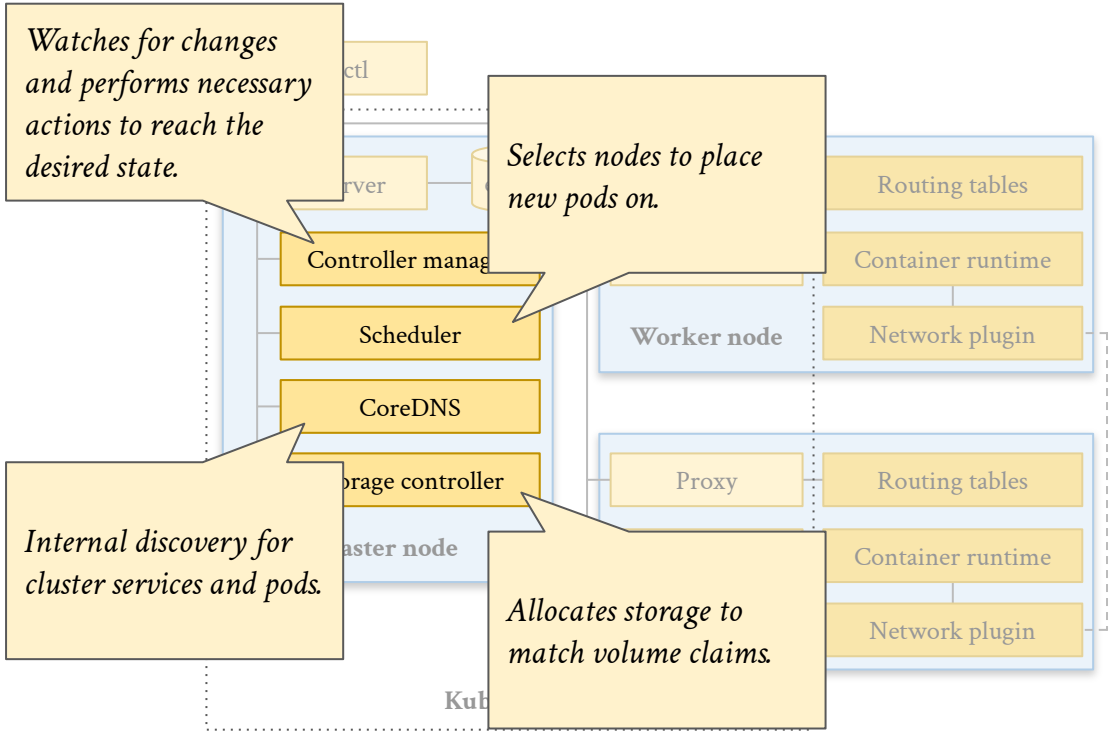

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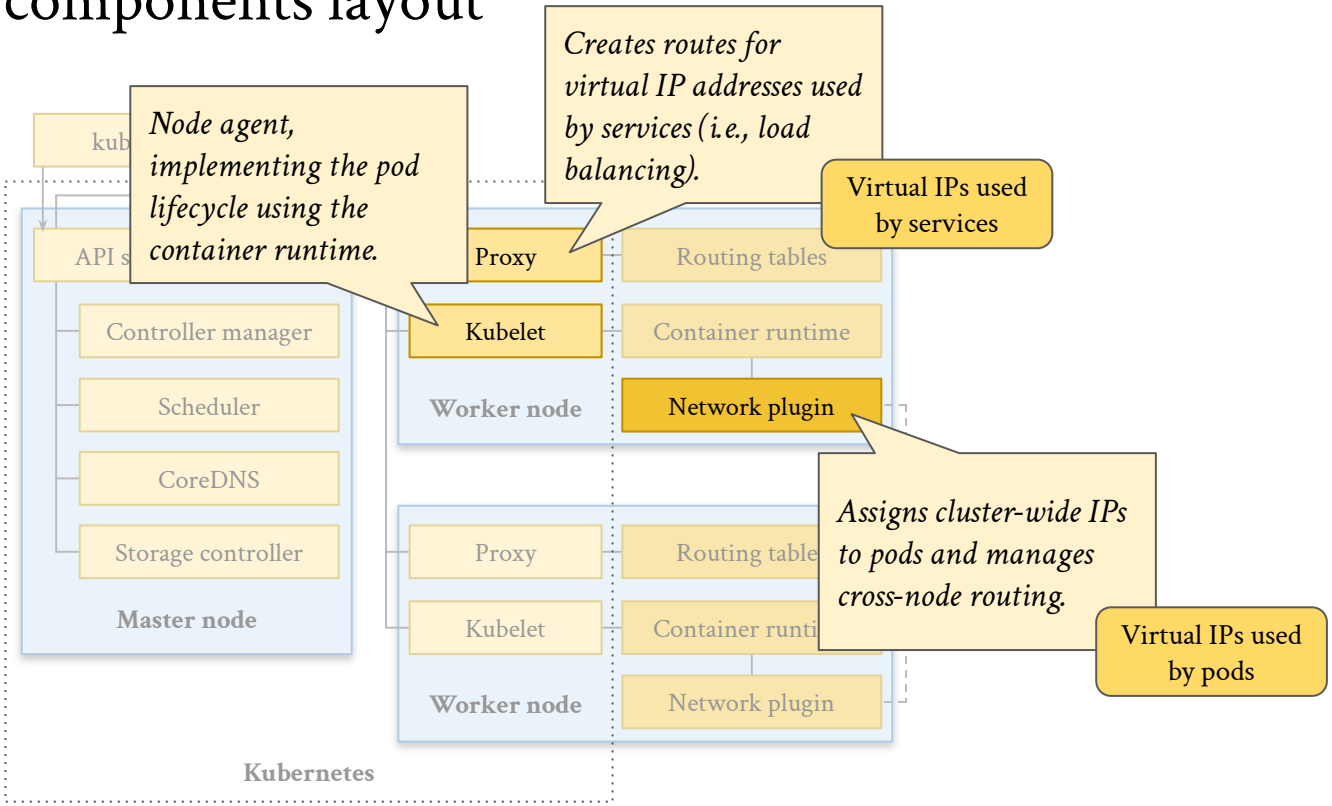

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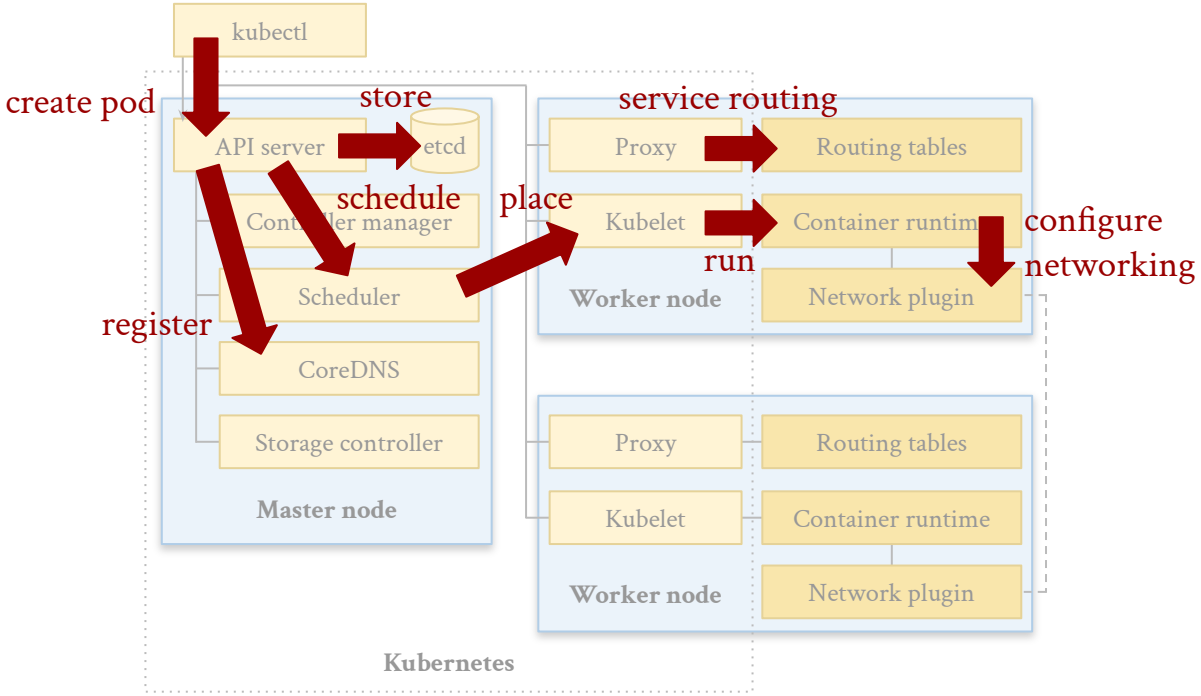
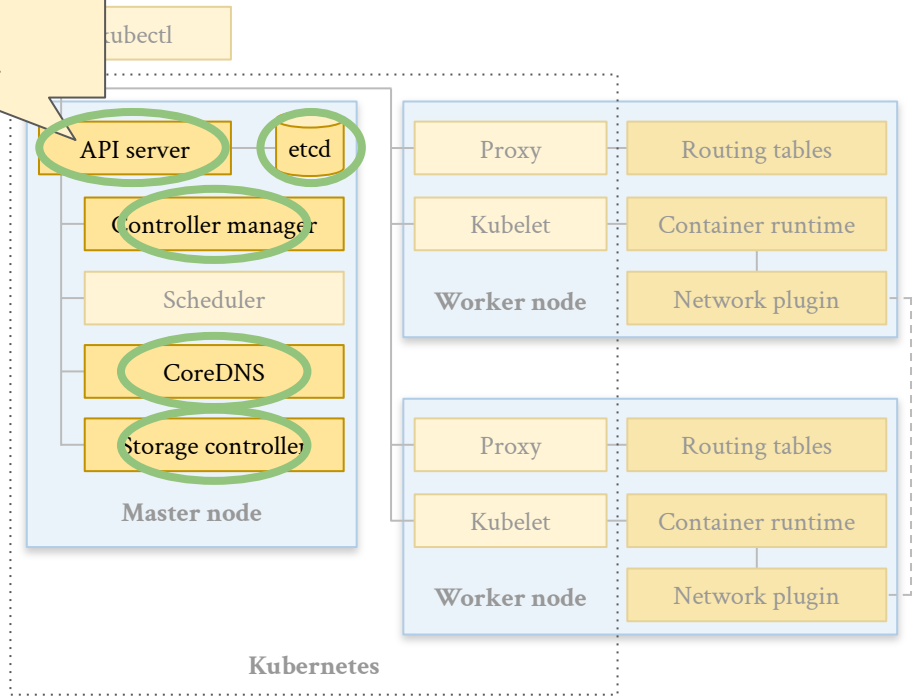
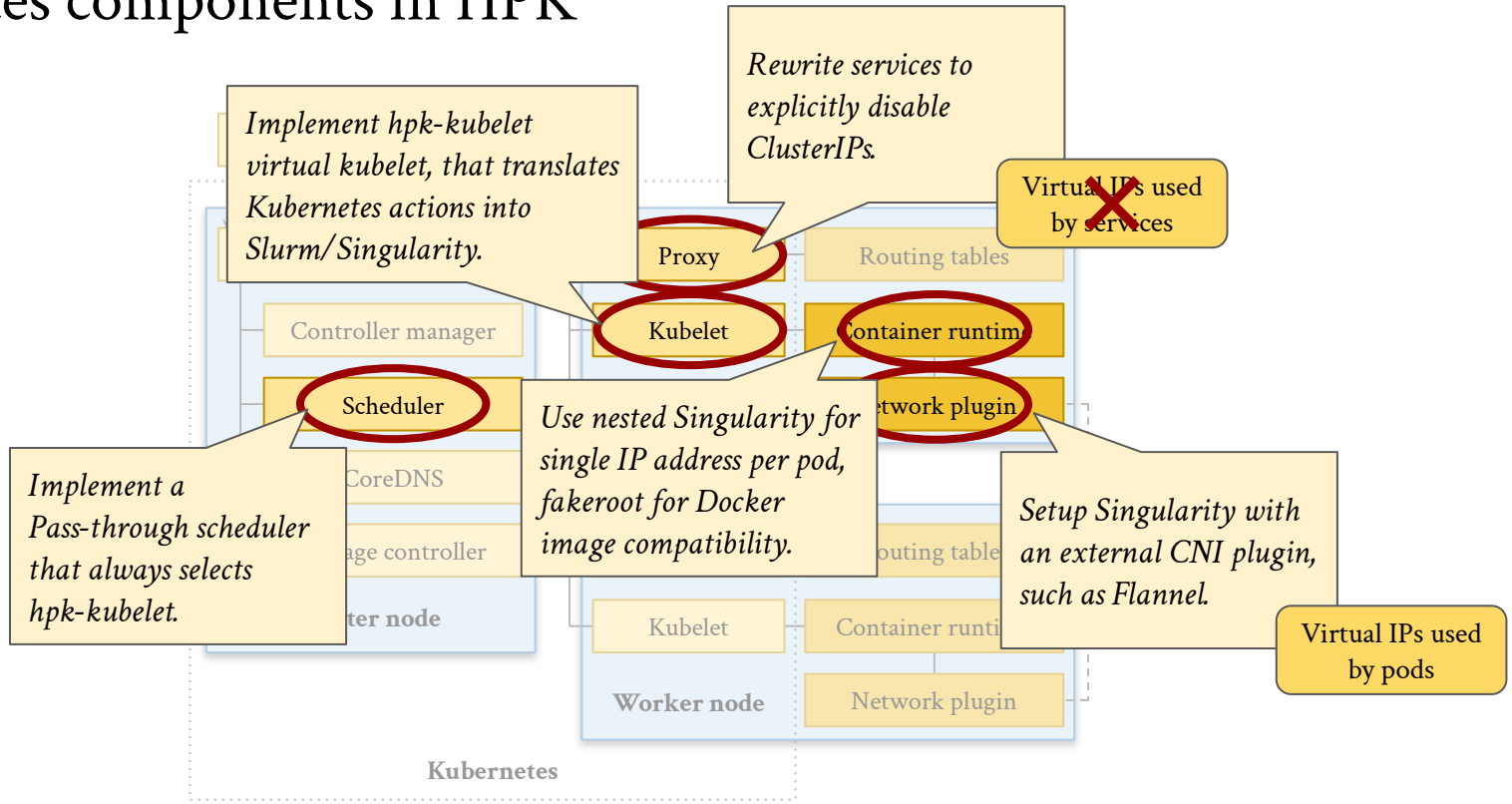


Kubernetes components in HPK

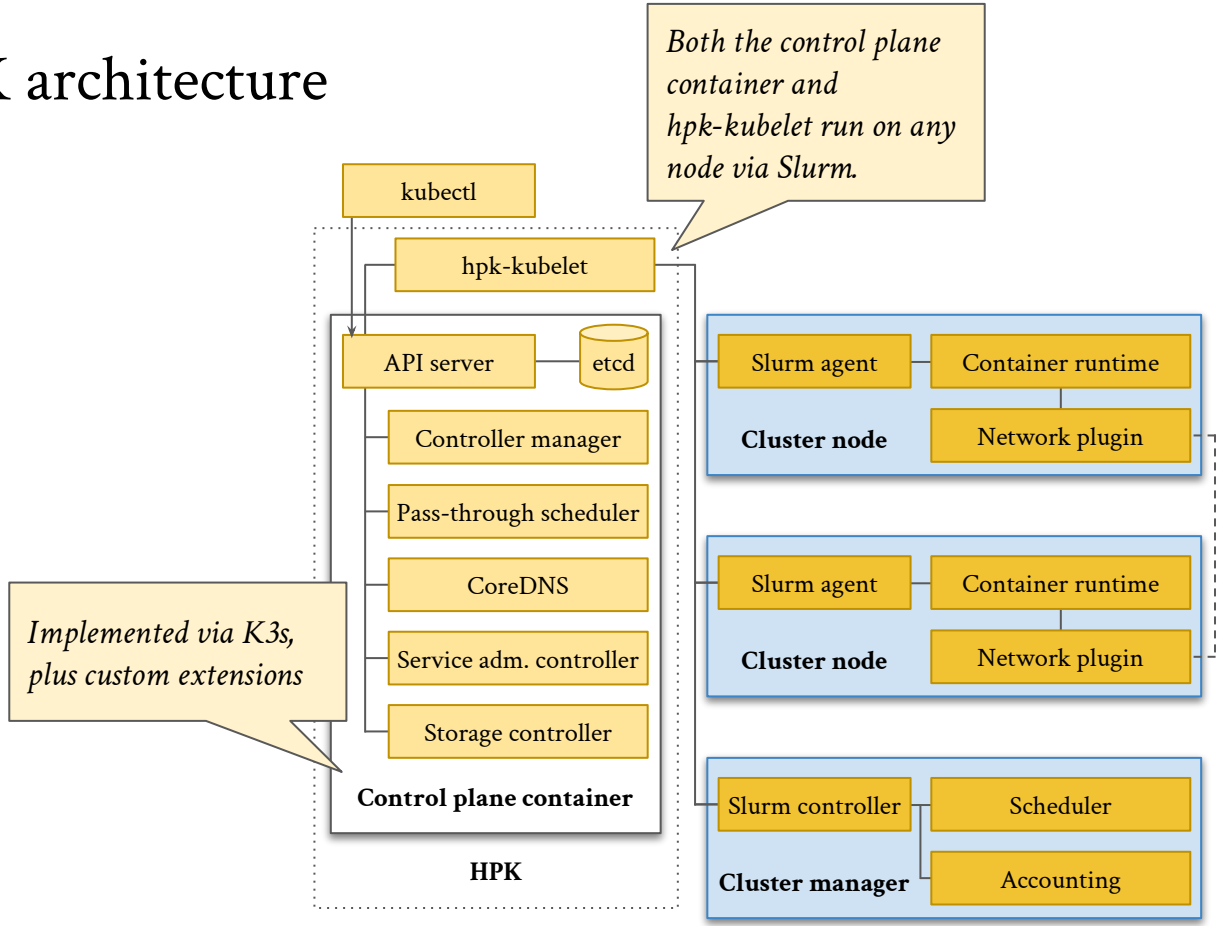
Services that run "out of the box", with no special dependencies, or minimal changes.



Kubernetes components in HPK



HPK architecture



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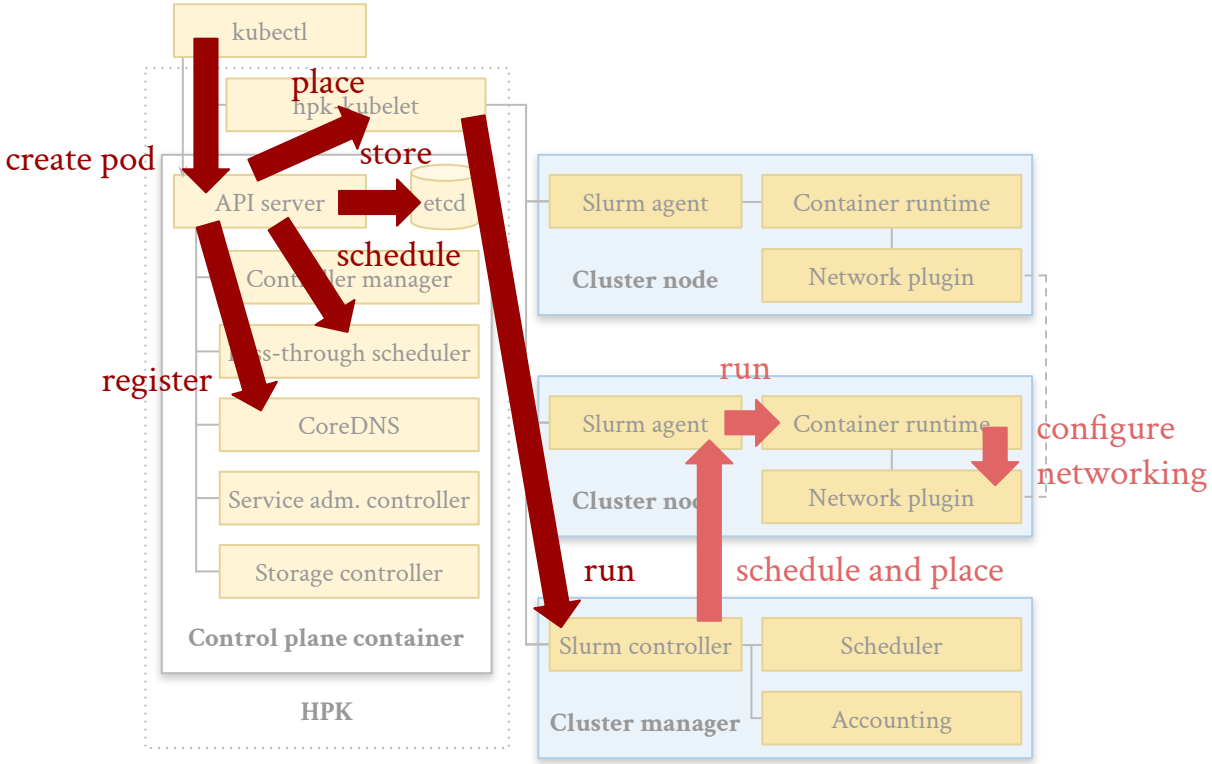
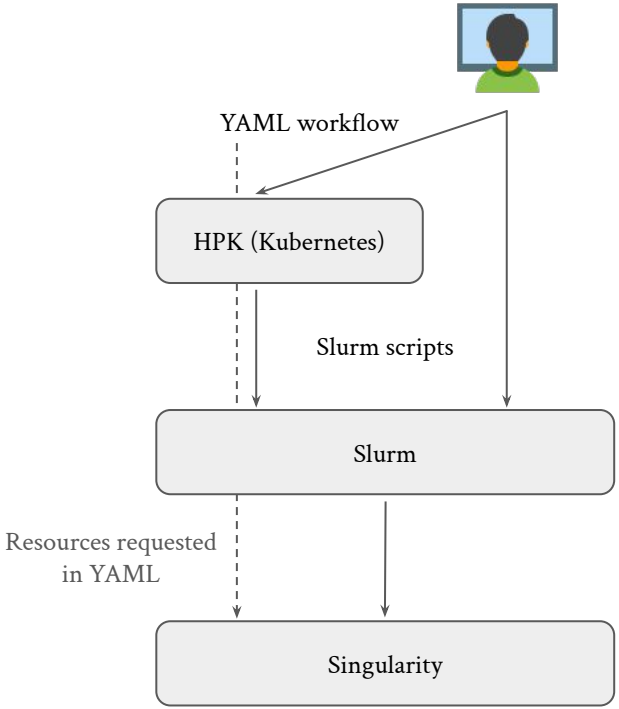


HPK implementation

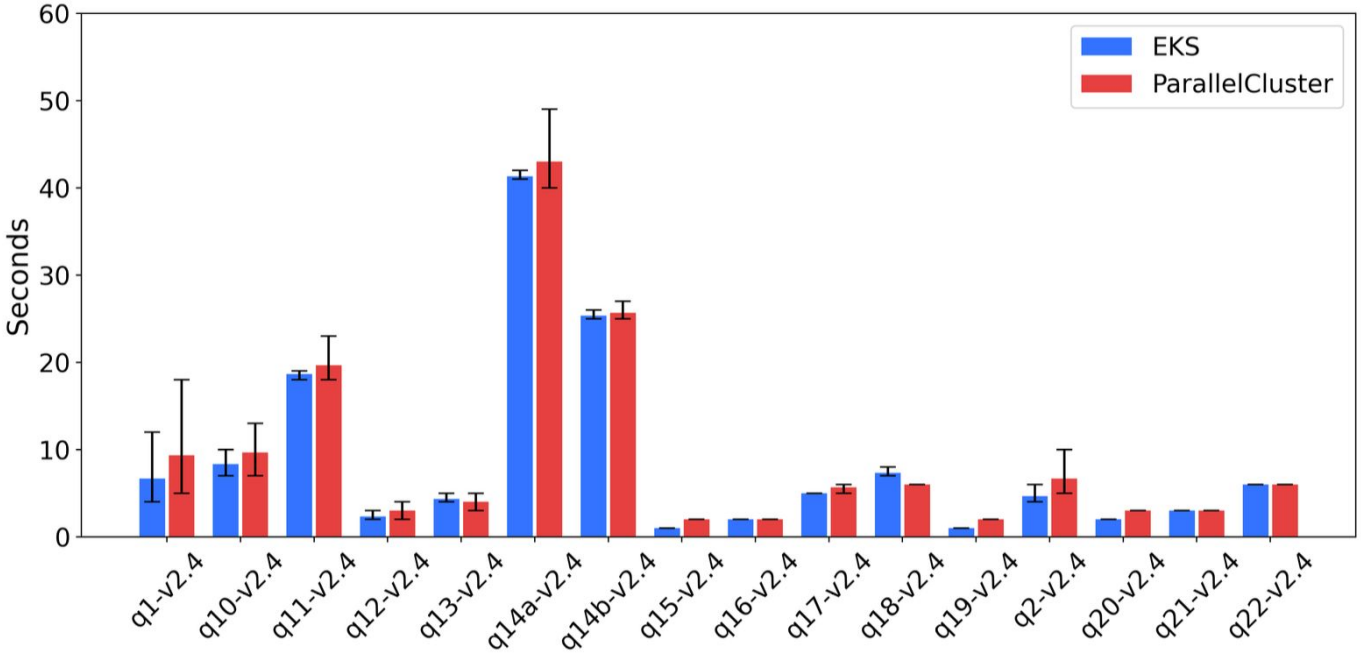
- HPK translates Kubernetes to Slurm scripts
  - Pods/jobs enter as YAML through the Kubernetes API, exit as Slurm scripts from hpk-kubelet → Pods show up as Slurm jobs
  - Slurm jobs run a hierarchy of Singularity containers (containers within pods) → Pause container is the pod
  - Kubernetes resource requirements end up in Slurm allocation requests → No changes to accounting
- HPK runs as a user process via Slurm
  - User can run both Kubernetes and Slurm workloads at the same time
  - No "special" allocation needed for HPK → 1 CPU, few GBs of RAM should be enough
  - Little support needed by the environment → No Slurm modifications, some Singularity networking configuration (inc. Flannel or other CNI)



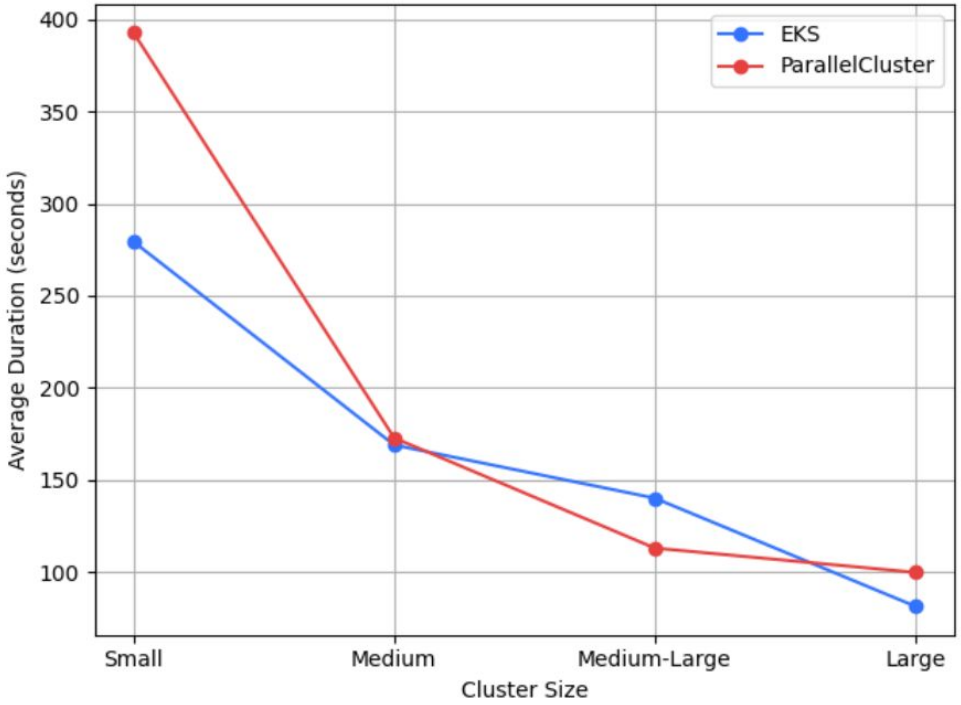
Evaluation

- Same applications on same hardware
  - EKS vs ParallelCluster on AWS
  - Applications deployed via Helm
  - Other software utilities may be necessary → MinIO
- Spark TCP-DS
  - Spark Operator → SparkApplication CRD
- Dask matrix multiplication
  - Varying configurations of driver/workers
- PyTorch model fine-tuning
  - Runs in a Jupyter notebook
  - Kubeflow Training Operator → PyTorchJob CRD
  - CPU-only vs GPU execution

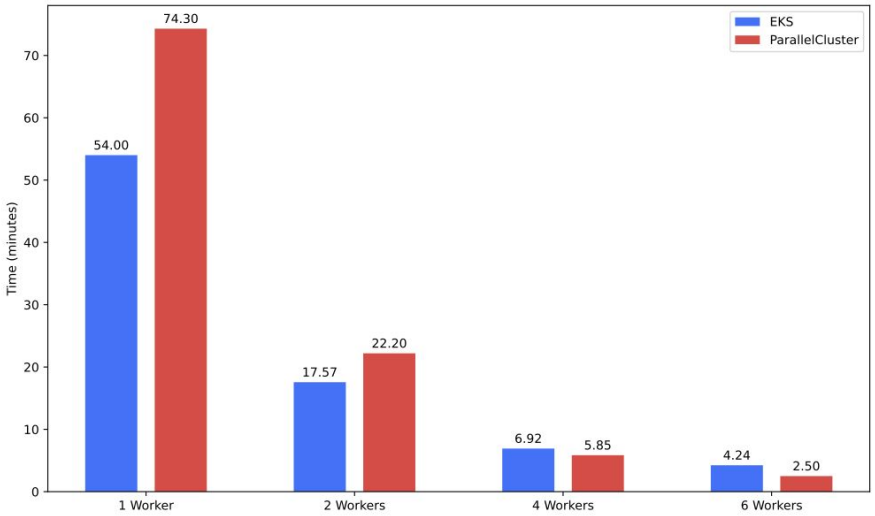
Spark TCP-DS


EKS
ParallelCluster
Seconds
0
10
20
30
40
50
60
q1-v2.4
q10-v2.4
q11-v2.4
q12-v2.4
q13-v2.4
q14a-v2.4
q14b-v2.4
q15-v2.4
q16-v2.4
q17-v2.4
q18-v2.4
q19-v2.4
q2-v2.4
q20-v2.4
q21-v2.4
q22-v2.4

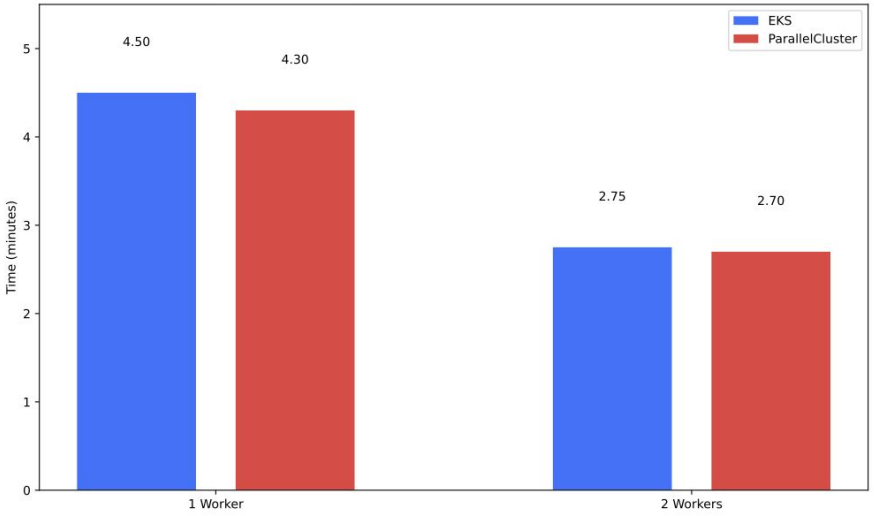
Dask


EKS
ParallelCluster
Average Duration (seconds)
400
350
300
250
200
150
100
Small
Medium
Medium-Large
Large
Cluster Size

PyTorch



CPU



GPU

Overview

- Running Kubernetes in Slurm is possible!
  - Currently working on integrating with large EuroHPC systems
  - Exploring options to avoid all system configuration requirements
  - Planning to contribute the project to HPSF
- HPK successfully deploys a variety of applications/utilities (also via Helm)
- Performance is similar to native Kubernetes (often even better!)
- Vision
  - Cloud-user PoV → Run on HPC hardware
  - HPC-user PoV → Exploit the Cloud software ecosystem
  - HPC centre PoV → Run Cloud workloads on the main HPC partition, attract users

Try it out! Feel free to contact us!

https://github.com/CARV-ICS-FORTH/HPK

Acknowledgements

DaFAB Riser grnet